

Ayyan Shahid

AI Engineer | Generative AI & LLM Systems | Agentic AI | Computer Vision

Lahore, Pakistan · Open to Remote & Relocation · +92 3316520087 ·

ayyanshahid640@gmail.com - linkedin.com/in/ayyan-shahid - github.com/Ayyan119

PROFESSIONAL SUMMARY

AI Engineer with close to 3 years of experience delivering production AI and software systems — the last 1.5 focused on Generative AI, Large Language Model (LLM) orchestration, and Computer Vision. Designs multi-agent pipelines with LangGraph and the Model Context Protocol (MCP), builds Retrieval-Augmented Generation (RAG) systems over vector databases, and ships end-to-end services on FastAPI, async PostgreSQL, and Amazon Web Services (AWS). Pairs a strong backend and full-stack foundation with applied computer vision — vehicle damage detection, image reimaging, and video pipelines — to turn models into reliable, real-world products.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Dart, SQL

Generative AI & LLMs: LLMs, LangGraph, LangChain, LangSmith, Retrieval-Augmented Generation (RAG), Corrective & Agentic RAG, Multi-Agent Systems, Model Context Protocol (MCP), Prompt Engineering, Tool Use, Memory & State Management, Fine-tuning (LoRA), Google Gemini, OpenAI

Vector DBs & Retrieval: Pinecone, ChromaDB, FAISS, Embeddings, Semantic & Hybrid Search

Computer Vision: PyTorch, TensorFlow, OpenCV, YOLOv8, Faster R-CNN, CNNs, Image Generation & Editing, Video Frame Processing, Segment Anything (SAM)

Backend & APIs: FastAPI, PostgreSQL, Microservices, Server-Sent Events (SSE), Pydantic

Cloud, MLOps & DevOps: Amazon Web Services (AWS — S3, EC2), Google Cloud / Vertex AI, Docker, Kubernetes, Continuous Integration / Continuous Deployment (CI/CD), GitHub Actions, MLflow, DVC, N8N

Frontend (supporting): Flutter, React.js

PROFESSIONAL EXPERIENCE

AI Engineer | Center for Advance Solutions

Dec 2025 – Present

- Developed AI applications using Python, LangChain, LangGraph, FastAPI, and Docker for real-world business solutions.
- Built RAG-based chatbots using vector databases and LLMs for document search and question answering.
- Developed computer vision models using OpenCV, TensorFlow, PyTorch, and YOLO for image classification and object detection.
- Deployed AI APIs, integrated machine learning models into applications, and collaborated using Git in an Agile environment.

Mobile App Developer | Enigmatix

Jan 2025 – Dec 2025

- Developed Android and Flutter applications for clients across education, healthcare, and business domains.
- Built mobile applications with Firebase, REST API integration, local databases, authentication, and responsive user interfaces.
- Collaborated with clients to develop, maintain, and enhance applications by implementing new features and resolving bugs.

Software Engineer (Android & Flutter) | Center for Advance Solutions

Aug 2023 - Dec 2024

- Developed native Android applications using Java, Android SDK, XML, and SQLite.
- Built cross-platform mobile applications for Android and iOS using Flutter and Dart.
- Integrated REST APIs, Firebase Authentication, Cloud Firestore, and local storage into production applications.
- Designed responsive user interfaces and implemented clean, maintainable code following industry best practices.
- Collaborated with backend developers using Git and Agile development methodologies.
- Fixed bugs, optimized application performance, and delivered feature enhancements based on client requirements.

PROJECTS

DocuMind AI — Enterprise RAG Document Assistant

Retrieval-Augmented Generation app that turns PDFs, contracts, and policies into a cited Q&A assistant. Async FastAPI backend with 3072-dimensional embeddings, hybrid semantic retrieval over Pinecone, conversational memory, confidence scoring, and real-time streaming via Server-Sent Events; Next.js + TypeScript front end.

Stack: LangChain · Pinecone · Gemini · FastAPI (async) · Next.js · TypeScript

BlogForge AI — Agentic Multi-Agent Writing System

Orchestrator-worker pipeline that plans, researches, writes, and illustrates long-form technical articles. A router classifies research mode, a Tavily agent gathers live sources, parallel worker agents draft sections, and a reducer subgraph merges output and generates images — with live, node-level streaming.

Stack: LangGraph · LangChain · Corrective RAG · Gemini · Tavily · Pydantic · Streamlit

TaleemAI — Adaptive Learning Platform (Urdu NLP)

EdTech platform built around a fine-tuned Urdu language model for multilingual content, a prerequisite knowledge graph spanning grade levels, a rules-based remediation engine, and a quiz and analytics layer for teachers.

Stack: LangChain · LangGraph · Fine-tuned Transformer (Urdu) · Knowledge Graph · FastAPI · Flutter

KnowledgeBase AI — Multi-Document Chatbot

Document intelligence platform that indexes PDFs, Word documents, and text files for conversational search. Supports semantic retrieval, conversation memory, and source-based answers.

Stack: LangChain · FAISS · Gemini · FastAPI · Streamlit

EDUCATION

B.Sc. in Computer Science

Islamia University of Bahawalpur · CGPA 3.87 / 4.0 · Ranked 1st in graduating batch